8. If $\vec{a}=\hat{i}-\hat{k}$, $\vec{b}=x\hat{i}+\hat{j}+(1-x)\hat{k}$, $\vec{c}=y\hat{i}+x\hat{j}+(1+x-y)\hat{k}$, show that $[\vec{a},\vec{b},\vec{c}]$ depends on neither x nor y.

9. If the vectors $a\hat{i}+a\hat{j}+c\hat{k}$, $\hat{i}+\hat{k}$ and $c\hat{i}+c\hat{j}+b\hat{k}$ are coplanar, prove that c is the geometric mean of a and b.

10. Let $\vec{a},\vec{b},\vec{c}$ be three non-zero vectors such that $\vec{c}$ is a unit vector perpendicular to both $\vec{a}$ and $\vec{b}$. If the angle between $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{6}$, show that $[\vec{a},\vec{b},\vec{c}]^2=\frac{1}{4}|\vec{a}|^2\,|\vec{b}|^2$.

6.5 Vector triple product

Definition 6.5

For a given set of three vectors $\vec{a},\vec{b},\vec{c}$, the vector $\vec{a}\times(\vec{b}\times\vec{c})$ is called a **vector triple product**.

Note

Given any three vectors $\vec{a},\vec{b},c$ the following are vector triple products :

$(\vec{a}\times\vec{b})\times\vec{c}$, $(\vec{b}\times\vec{c})\times\vec{a}$, $(\vec{c}\times\vec{a})\times\vec{b}$, $\vec{c}\times(\vec{a}\times\vec{b})$, $\vec{a}\times(\vec{b}\times\vec{c})$, $\vec{b}\times(\vec{c}\times\vec{a})$

Using the well known properties of the vector product, we get the following theorem.

Theorem 6.7

The vector triple product satisfies the following properties.

(1) $(\vec{a}_1+\vec{a}_2)\times(\vec{b}\times\vec{c}) = \vec{a}_1\times(\vec{b}\times\vec{c})+\vec{a}_2\times(\vec{b}\times\vec{c})$, $(\lambda\vec{a})\times(\vec{b}\times\vec{c})=\lambda(\vec{a}\times(\vec{b}\times\vec{c}))$, $\lambda\in\mathbb{R}$

(2) $\vec{a}\times((\vec{b}_1+\vec{b}_2)\times\vec{c}) = \vec{a}\times(\vec{b}_1\times\vec{c})+\vec{a}\times(\vec{b}_2\times\vec{c})$, $\vec{a}\times((\lambda\vec{b})\times\vec{c})=\lambda(\vec{a}\times(\vec{b}\times\vec{c}))$, $\lambda\in\mathbb{R}$

(3) $\vec{a}\times(\vec{b}\times(\vec{c}_1+\vec{c}_2)) = \vec{a}\times(\vec{b}\times\vec{c}_1)+\vec{a}\times(\vec{b}\times\vec{c}_2)$, $\vec{a}\times(\vec{b}\times(\lambda\vec{c}))=\lambda(\vec{a}\times(\vec{b}\times\vec{c}))$, $\lambda\in\mathbb{R}$

Remark

Vector triple product is not associative. This means that $\vec{a}\times(\vec{b}\times\vec{c})\neq(\vec{a}\times\vec{b})\times\vec{c}$, for some vectors $\vec{a},\vec{b},\vec{c}$.

Justification

We take $\vec{a}=\hat{i}$, $\vec{b}=\hat{i}$, $\vec{c}=\hat{j}$. Then, $\vec{a}\times(\vec{b}\times\vec{c})=\hat{i}\times(\hat{i}\times\hat{j})=\hat{i}\times\hat{k}=-\hat{j}$ but $(\hat{i}\times\hat{i})\times\hat{j}=\vec{0}\times\hat{j}=\vec{0}$.

Therefore, $\vec{a}\times(\vec{b}\times\vec{c})\neq(\vec{a}\times\vec{b})\times\vec{c}$.

The following theorem gives a simple formula to evaluate the vector triple product.

Theorem 6.8 (Vector Triple product expansion)

For any three vectors $\vec{a},\vec{b},\vec{c}$ we have $\vec{a}\times(\vec{b}\times\vec{c})=(\vec{a}\cdot\vec{c})\vec{b}-(\vec{a}\cdot\vec{b})\vec{c}$.

Proof

Let us choose the coordinate axes as follows :

Let x-axis be chosen along the line of action of $\vec{a}$, y-axis be chosen in the plane passing through $\vec{a}$ and parallel to $\vec{b}$, and z-axis be chosen perpendicular to the plane containing $\vec{a}$ and $\vec{b}$. Then, we have

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